

TABLE I
FIXED CAPITAL

LAND AND BUILDING			
Description	Details		
Built-up area	300sqm		
Office and stores	50sqm		
Factory shed (workshop, assembling and testing)	250sqm		
Rent payable (₹ per annum)	60,000		
MACHINERY AND EQUIPMENT			
Particulars	Indian/Imported	Quantity	Amount (₹)
(a) Machinery			
CNC winding machine	Indian	2	100,000
Semi-automatic winding machine	Indian	1	35,000
Oven (3.5kW)	Indian	1	30,000
Vacuum impregnation plant (30 litres capacity)	Indian	1	80,000
(b) Testing equipment			
Digital LCR-Q meter	Indian	1	13,000
Oscilloscope (0MHz-20MHz)	Indian	1	20,000
Insulation tester	Indian	1	6,000
High-voltage break-down tester	Indian	2	15,000
Megger	Indian	1	4,000
4.5-digit digital multimeter	Indian	1	8,500
Analogue multimeter	Indian	1	8,000
Total			319,500
Electrification @ 10% of total cost of machinery and equipment			31,950
Tools, jigs, fixtures and soldering iron station			45,000
Office equipment and furniture			50,000
Pre-operative expenses			10,000
Total			456,450

Testing. Transformers are individually-tested. Following routine tests are conducted in the lab:

1. No load loss
2. Full load loss
3. Double frequency
4. Double voltage
5. High voltage
6. Insulation
7. Measurement of resistance
8. Heat run
9. Break down voltage (BDV)

Good manufacturing practices

Pollution control. The government places utmost importance on controlling environmental pollution. Therefore small-scale entrepreneurs need to follow environment-friendly practices and adopt pollution control measures by process modification and technology substitution.

India acceded to Montreal Protocol in September 1992. Hence, production and use of ozone-depleting substances like chlorofluoro carbon (CFC), carbon tetrachloride, halons and methyl

chloroform must be phased out immediately with alternative chemicals/solvents. A notification for detailed rules to regulate ozone-depleting substances phase-out under Environment Protection Act, 1986 was put in place with effect from July 19, 2000.

Energy conservation. With growing energy needs and shortage coupled with rising energy costs, a greater thrust in energy efficiency in the industrial sector has been given by the government of India since 1980s. The Energy Conservation Act, 2001 was enacted on August 18, 2001. It provides for efficient use of energy, its conservation and capacity building.

Following steps may help in the conservation of electrical energy:

1. Adoption of energy-conserving technologies, production aids and testing facilities
2. Efficient management of processes/manufacturing machineries and systems, quality control and testing equipment for yielding maximum energy conservation

3. Using efficient temperature-controlled soldering and de-soldering stations for optimum use of electrical energy for heating during soldering

4. Periodical maintenance of motors, compressors and the like

5. Use of power factor correction capacitors, proper selection and layout of lighting system, timely switching on/off of lights, use of compact fluorescent lamps wherever possible and so on

Raw materials and equipment

Basic raw materials for making a transformer include laminated core, windings and insulating materials.

As a manufacturer, you must procure some basic machinery and equipment for this business, including the following: CNC winding machine, semi-automatic winding machine, oven (3.5kW), vacuum impregnation plant, digital LCR-Q meter, oscilloscope, insulation tester, high-voltage break-down tester, megger, 4.5-digit digital multimeter, analogue multimeter, and tools, jigs, fixtures and soldering iron station.

In addition, arrange utilities like water and electricity.

Further, a medium- and large-scale unit demands 300sqm area for the operation.

Business economics

An indicative analysis has been given on the following basis and presumptions:

1. Basis for calculation of production capacity is taken on a single shift of eight hours per day basis on 75 per cent efficiency.

2. Maximum capacity utilisation on single-shift basis for 300 days a year is considered. During the first and second years of operations, capacity utilisation is 60 and 80 per cent, respectively. The unit is expected to achieve full capacity utilisation from third year onwards.

3. Salaries and wages, cost of raw materials, utilities, civil construction, etc are based on prevailing rates in and around north India. These cost factors are likely to vary with time and location.

4. Interest on term and working capital loans is taken at the rate of 16 per cent on average. This rate may vary depending on the policy of the financial institution/agency from time to time.

5. Cost of machinery and equipment refer to a particular make/model.